DATA11001 DATA SCIENCE:

MINIPROJECT

THE REPORT

Analysis of AI's Effect on Unemployment for Policymakers

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1 Introduction

The use of artificial intelligence has increased rapidly during the past three years. It has been harnessed for both entertainment and practical purposes in a wide variety of ways. It has, in fact, become a part of our everyday lives surprisingly quickly. Often, when such new technology is adopted rapidly, the downsides of the technology are overlooked. We do not yet know the effects that this will have on our society. As the use of artificial intelligence has become widespread, countries have awakened to its impact on areas such as the labour market and employment.

This report examines the relationship between enterprise AI adoption and unemployment in Finland. It combines an analysis of AI adoption and registered unemployment with an examination of historical seasonal patterns and annual minimum unemployment levels. The aim is to describe the available evidence and its limitations.

2 Data

The following three datasets were used: Statistics Finland occupation specific unemployment data (categorized by ISCO) [4], Statistics Finland industry specific AI adoption data (categorized by NACE) [5] and Eurostat employment data (used to recategorize AI adoption data from NACE to ISCO) [1].

The unemployment dataset contains counts of unemployed job seekers registered at the end of each month, categorized by ISCO 4-digit occupation codes. For the purposes of comparing AI adoption data and unemployment data, the 4-digit ISCO codes were aggregated into 1-digit groups. OC0 (Armed forces occupations) and OC6 (Skilled agricultural, forestry and fishery workers) were excluded from the dataset, as comparison with AI adoption was not feasible for these occupation groups due to a lack of data. The AI adoption data is categorized by industry (NACE) instead of occupation (ISCO), so an estimated mapping from NACE to ISCO was used. The mapping used the Eurostat employment data for weighting.

3 Results

3.1 Correlation analysis and prediction

Figure 1 plots the percentage of companies using AI alongside the annual unemployment count for all occupation groups. It helps us see whether unemployment rises or falls

while AI adoption increases. For example, if AI usage increases while unemployment also increases, this could suggest a possible relationship worth investigating further.

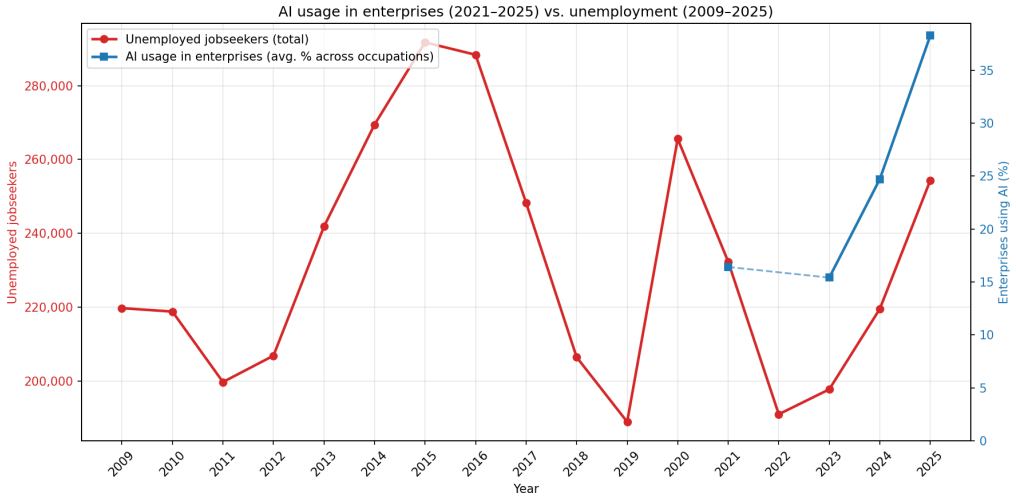


Figure 1: Artificial intelligence adoption and unemployment count accross all occupations

The plot shows that unemployment is rising alongside AI adoption in enterprises; however, there is not enough data to determine if this is a long-term trend.

Figure 2 looks at different occupation groups, such as managers, professionals, service workers and construction workers separately. It compares their AI adoption percentages with their unemployment counts and changes over time. Unemployment is indexed to 2021, meaning that each group's unemployment level in 2021 is set to 100. This makes it easier to compare whether unemployment has increased or decreased proportionally in different groups, even if their original numbers were very different. The second part of the plot compares changes in AI adoption with percentage changes in unemployment between 2023 and 2025.

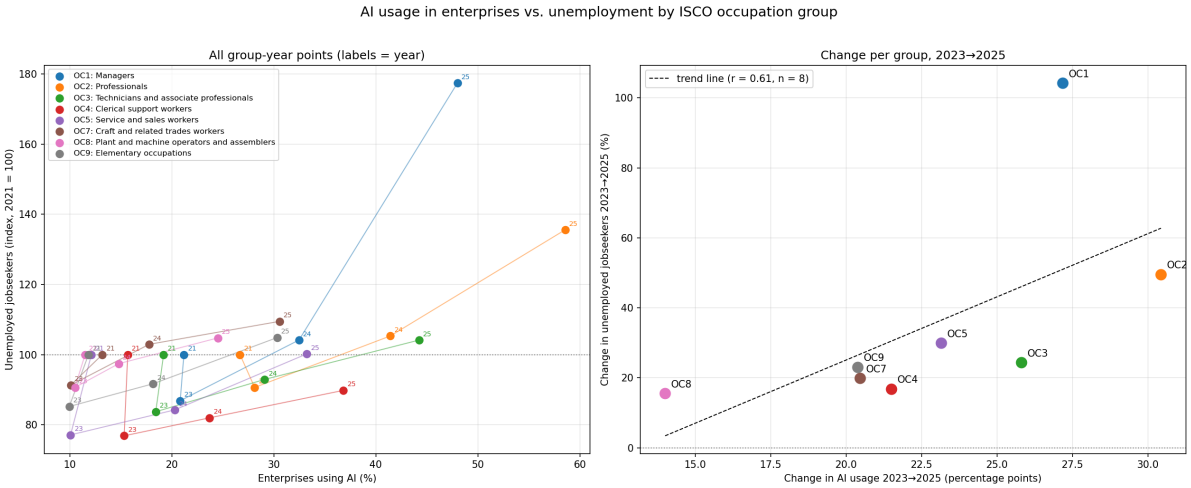


Figure 2: Artificial intelligence and unemployment scatter plot

Generative AI is expected to impact knowledge work, such as programming, management and customer support more strongly than physical work [2] [3]. This is due to the fact that generative AI is limited in its ability to interact with the physical world, so professions requiring physical labour are less likely to be impacted. The plot shows a stronger positive association between AI adoption and unemployment for OC1 (Managers) and OC2 (Professionals), which are more knowledge work heavy groups. However, the plot also shows that OC4 (Clerical support workers), which consists of repetitive knowledge work and is considered highly exposed to generative AI, has the weakest association across all groups. In light of this and the lack of data, it cannot be concluded with sufficient certainty that AI adoption is reliably correlated with the rise in unemployment.

Figure 3 shows predicted unemployment. The model considers AI adoption, occupation group and year when estimating unemployment. The red dashed line represents the unemployment forecast, while the shaded area is an 80% prediction band. The blue dashed curves represent AI adoption scenarios with ceilings of 70%, 85% and 100%. These ceilings are assumptions rather than observed outcomes. The prediction band in Figure 3 is quite large, which reflects the lack of data.

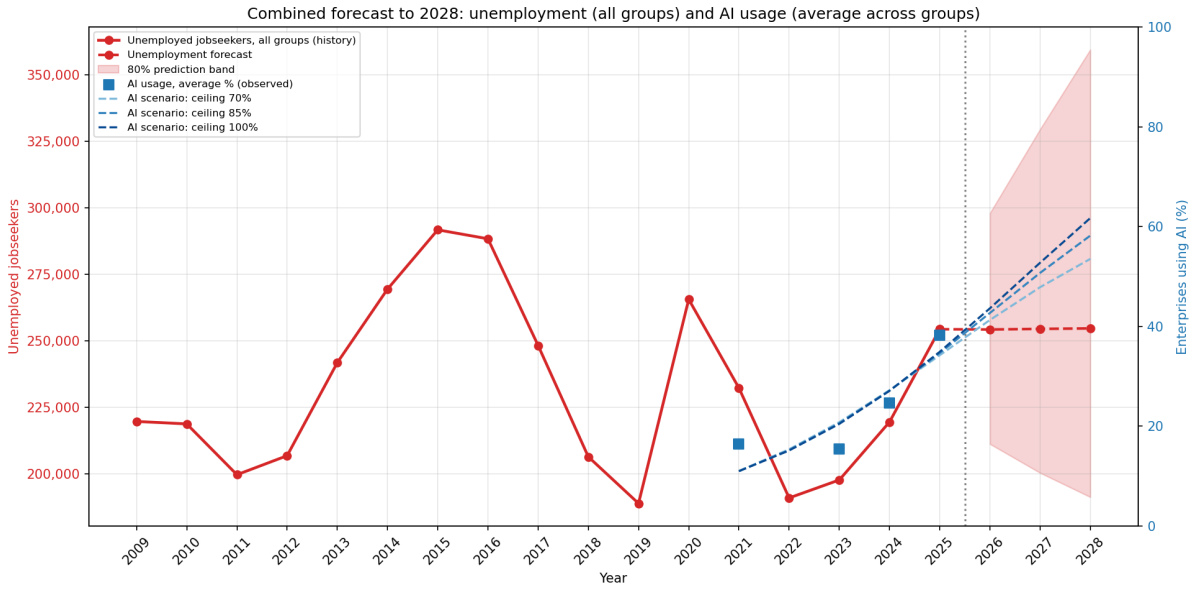


Figure 3: Unemployment prediction

The graphs had several limitations that made it difficult to draw firm conclusions about the relationship between AI adoption and unemployment. Correlation alone cannot establish causation. The occupational scatter plot provided some detail, but the small number of groups and the relatively short time period limited the reliability of the comparisons. Additionally, adoption of AI in enterprises does not necessarily reflect how AI affects individual occupations.

3.2 Seasonal cycle and unemployment floor analysis

Figure 4 shows seasonal cycles in unemployment for each occupation group. A seasonal cycle is repeating pattern, where the studied signal rises and falls in similar ways during similar times along multiple years. Seasonal cycles are useful for detecting abrupt disturbances in the job market. This is useful in the context of analysing if AI has had an impact on the job market, as AI is a very disruptive technology that has the potential to completely eliminate certain professions. If AI is truly having a dramatic impact, it should register as a disturbance in the seasonal cycle.

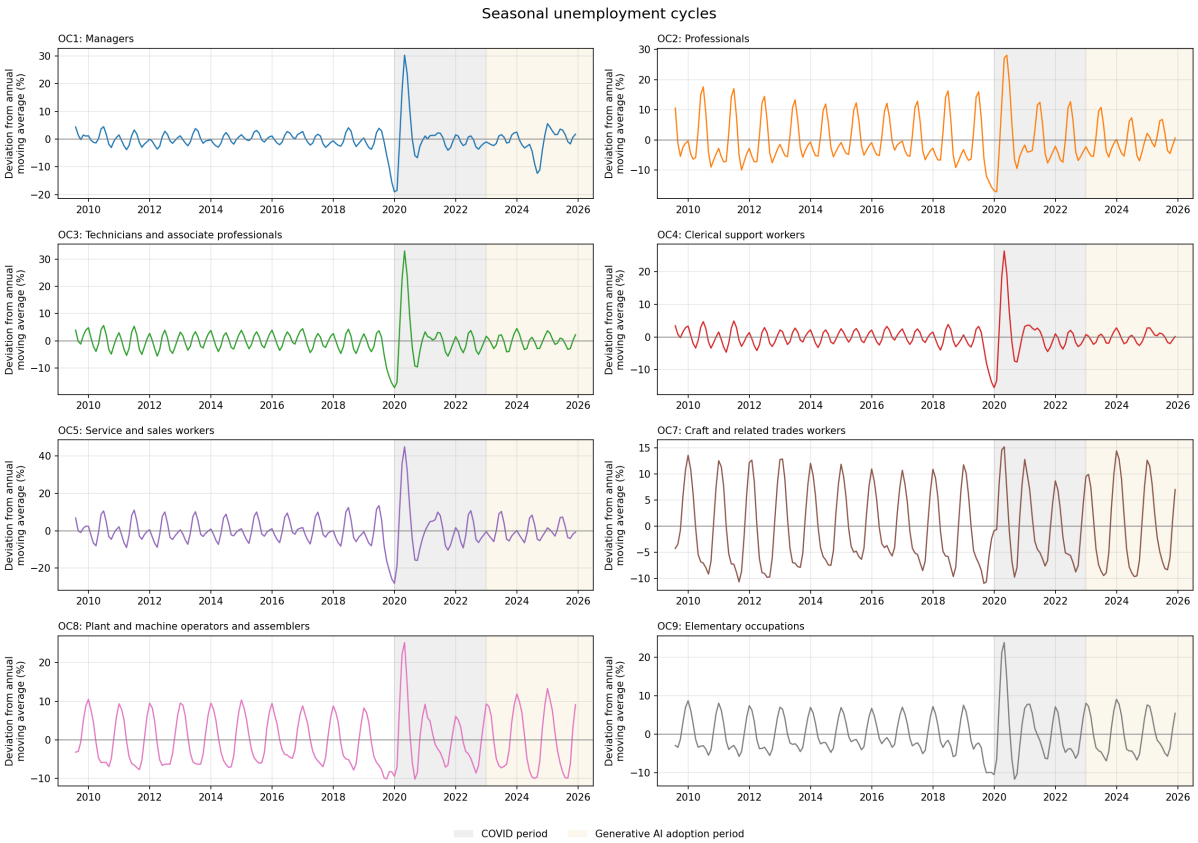


Figure 4: Seasonal unemployment cycles

The plot clearly shows the disruption caused by the COVID pandemic for all occupation groups except OC7 (Craft and related trades workers). However, the cycle for all groups continues repeating as usual post-COVID and into the generative AI adoption period. The only noticeable anomaly is the larger than usual negative spike that occurs for OC1 around the second half of 2024. This suggests that there haven't been any dramatic/COVID-level disturbances to the job market during the generative AI adoption period.

Figure 5 shows the minimum annual unemployment for all occupation groups indexed to the level of 2019, which is set to 100. A rise in the minimum annual unemployment indicates that the job market is getting worse. If AI is lessening the overall need for workers, the minimum annual unemployment should keep rising past the historical baseline.

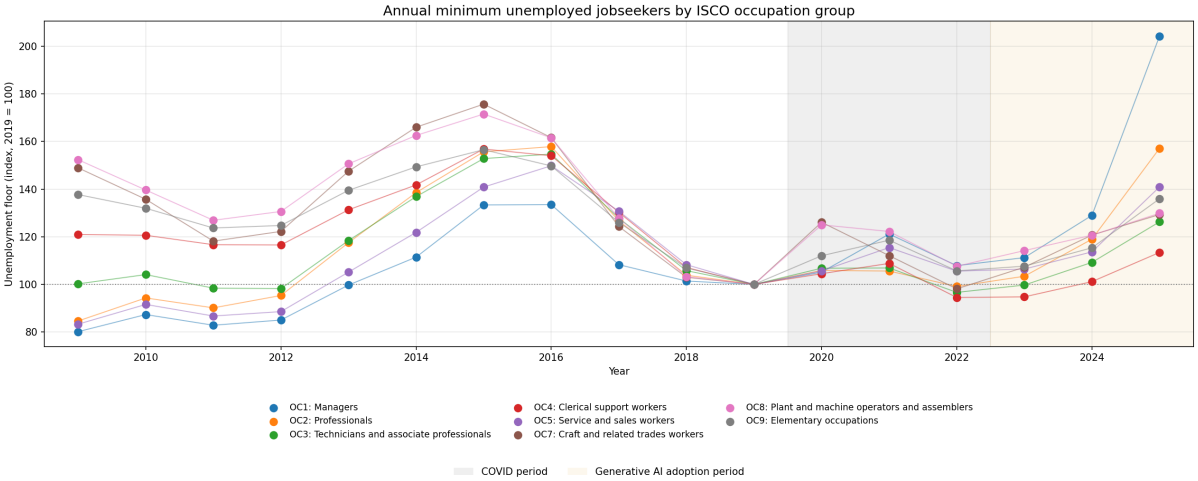


Figure 5: Unemployment floor

The plot shows that the minimum annual unemployment is rising during the generative AI adoption period. However, there is not enough data to determine if this is a long-term trend. The minimum annual unemployment could settle at levels around the 2015 peak or continue rising.

4 Conclusion

Unemployment has been rising, but this analysis shows no evidence of a causal link with AI adoption. There is limited evidence that AI may not be having a big (COVID-level or higher) impact on unemployment yet. This is due to the fact that unemployment for occupation groups that would theoretically be impacted most by AI is not reliably correlated with AI adoption and the job market has not seen a disruption to its typical seasonal cycles.

Lack of data on AI adoption makes it hard to evaluate how AI is affecting work and hiring. The best data currently available for Finland reports AI adoption percentages in different industries, but there is no detailed data on, for example, how much more productive AI makes a worker in a specific profession. There is also the fact that AI is an international phenomenon, so its impacts should not be examined solely on a country-by-country basis. Consequently, as a direction for further development, AI adoption data and unemployment data should be collected across multiple countries and more detailed data should be used if available.

Sources

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The authors employed ChatGPT and Gemini as supportive tools for language polishing and statistical code generation. All AI-assisted contributions were independently verified and edited by the author, who bears complete responsibility for the analytical results and conclusions contained in this study.